

12.70

EE25BTECH11057 - Rushil Shanmukha Srinivas

Question: X bullocks and Y tractors take 8 days to plough a field. If we halve the number of bullocks and double the number of tractors, it takes 5 days to plough the same field. How many days will it take X bullocks alone to plough the field?

Solution:

The total work done = W

Work done by 1 bullock in one day = x

Work done by 1 tractor in one day = y

Given,

$$\begin{pmatrix} 8X & 8Y \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = W \quad (0.1)$$

$$\begin{pmatrix} \frac{5X}{2} & 10Y \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = W \quad (0.2)$$

The combined matrix is

$$\begin{pmatrix} 8X & 8Y \\ \frac{5X}{2} & 10Y \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} W \\ W \end{pmatrix} \quad (0.3)$$

Forming the augmented matrix ,

$$\left(\begin{array}{cc|c} 8X & 8Y & W \\ \frac{5X}{2} & 10Y & W \end{array} \right) \quad (0.4)$$

Using Gaussian Eliminations

$$\left(\begin{array}{cc|c} 8X & 8Y & W \\ \frac{5X}{2} & 10Y & W \end{array} \right) \xrightarrow{R_2 \rightarrow 4R_2 - 5R_1} \left(\begin{array}{cc|c} 8X & 8Y & W \\ -30X & 0 & -W \end{array} \right) \quad (0.5)$$

$$30Xx = W \quad (0.6)$$

Number of days taken = 30